SYMMETRIC CYCLOTOMIC HECKE ALGEBRAS

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In this paper we prove that the generic cyclotomic Hecke algebras for imprimitive complex reflection groups are symmetric over any ring containing inverses of the parameters. For this we show that the determinant of the Gram matrix of a certain canonical symmetrizing form introduced in [**3**] is a unit in any such ring. On the way we show that the Ariki-Koike bases of these algebras are also quasi-symmetric.

0. Introduction and statement of the main result

Work of Broué, Michel and the first author on ℓ-blocks in finite groups of Lie type suggest that the decomposition of certain Lusztig induced characters should be governed by suitable complex reflection groups. This lead Broué and the first author in [**4**] to introduce the concept of the cyclotomic Hecke algebra $\mathcal{H}(W,\mathbf{u})$ associated to a complex reflection group W. Conjecturally specializations of this algebra should occur as endomorphism algebras of Lusztig induced characters, thus explaining the aforementioned observation (see [**4**]).

This general conjecture has been proved only in a small number of cases. Nevertheless, several structural and numerical consequences of this conjecture have been verified. One of the properties of cyclotomic algebras implied by the general conjecture is that $\mathcal{H}$ should carry a canonical symmetrizing form making it into a symmetric algebra over the ground ring A in which just the parameters are inverted. Such a canonical symmetrizing form over the field of fractions of A was constructed in [**3**] for all but finitely many of the cyclotomic algebras associated to irreducible complex reflection groups. In this paper we prove that pairs of dual bases with respect to this form are defined over A:

Theorem. *The generic cyclotomic algebra $\mathcal{H}=\mathcal{H}(W,\mathbf{u})$ of an irreducible imprimitive complex reflection group W is symmetric over $\mathbb{Z}[\mathbf{u},\mathbf{u}^{-1}]$.*

The proof of this result will be given in the subsequent sections. There we will also give more precise statements about properties of the corresponding symmetrizing form.

The assertion of the main theorem is well known in the case of Iwahori-Hecke algebras, *i.e.*, for cyclotomic algebras for real reflection groups. In that case a dual basis can immediately be written down in terms of the standard basis of $\mathcal{H}$ and the

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assertion follows by inspection. It turns out that the situation for the complex case is more subtle.

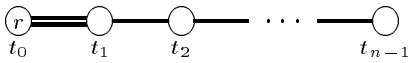
We would like to point out that it is still open whether the cyclotomic algebras for all of the finitely many primitive complex reflection groups enjoy the properties stated in the theorem.

1. The symmetrizing form on $\mathcal{H}(W_n, \mathbf{u})$

In this section we recall the definition of the cyclotomic Hecke algebra $\mathcal{H}(W_n, \mathbf{u})$ and the construction of a symmetrizing form on $\mathcal{H}(W_n, \mathbf{u}) \otimes \mathbb{Q}(\mathbf{u})$ from [**3**].

Let V be the n-dimensional unitary space $V = \langle e_1, \dots, e_n \rangle$ with the standard unitary inner product. Fix $r > 1$, and let $W_n = G(r, 1, n) \subset \mathrm{GL}(V)$ be the imprimitive complex reflection group on V generated by the reflections $t_1, \dots, t_{n-1}$ of order 2 with roots $e_2 - e_1, \dots, e_n - e_{n-1}$, respectively, and a (complex) reflection t_0 of order r with root e_1. On these generators, W_n has a presentation given by the diagram

(1.1) $B_n^{(r)}$:



The elements $t_1, \dots, t_{n-1}$ generate a subgroup of W_n isomorphic to the symmetric group $\mathfrak{S}_n$, and form a set of standard generators of this Coxeter group. We will consider $\mathfrak{S}_n$ as a subgroup of W_n via this embedding.

Let $\mathbf{u} = (u_1, \dots, u_r, q)$ be indeterminates and $A := \mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$, where we write $\mathbf{u}^{-1} := (u_1^{-1}, \dots, u_r^{-1}, q^{-1})$. The *generic cyclotomic algebra* $\mathcal{H} := \mathcal{H}(W_n, \mathbf{u})$ attached to W_n is the free unital associative A-algebra on generators $T_0, T_1, \dots, T_{n-1}$, subject to the braid relations implied by the diagram (1.1) and the deformed order relations

$$(T_0 - u_1) \dots (T_0 - u_r) = 0 = (T_i - q)(T_i + 1), \qquad 1 \le i \le n-1,$$

(see [**4**, 4.1]). Under the specialization

(1.2) $$u_k \mapsto \zeta_r^k, \quad 1 \le k \le r, \qquad q \mapsto 1,$$

where $\zeta_r := \exp(2\pi \mathrm{i}/r)$, the cyclotomic algebra specializes to the group algebra of W_n over $\mathbb{Z}[\zeta_r]$. Further, it is known that $\mathcal{H}$ is a free A-algebra of rank $|W_n|$ (see [**2**]). Thus the subalgebra generated by $T_1, \dots, T_{n-1}$ can and will be identified with the generic Iwahori-Hecke algebra $\mathcal{H}(\mathfrak{S}_n, q)$ of type A_{n-1}.

For $\mathbf{x} = (x_1, \dots, x_l)$, $x_i \in \{t_0, t_1, \dots, t_{n-1}\}$, a reduced expression of an element $x = x_1 \dots x_l$ in W_n, let $T_{\mathbf{x}} := T_{x_1} \dots T_{x_l}$. Note that it is *not* in general true that two reduced expressions $\mathbf{x}, \mathbf{x}'$ of the same element $x \in W_n$ give rise to identical elements $T_{\mathbf{x}}, T_{\mathbf{x}'}$ in $\mathcal{H}$. Nevertheless, this property does hold if $x \in \mathfrak{S}_n \le W_n$.

Let $X = \{\mathbf{x} \mid x \in W\}$ be an arbitrary set of reduced expressions for the elements of W. Then it is known that the set $B_X := \{T_{\mathbf{x}} \mid \mathbf{x} \in X\}$ is an A-basis of $\mathcal{H}$ (see for example [**3**, Prop. 2.4]). Any such basis will be called a *reduced basis of* $\mathcal{H}$. Choose one such reduced basis B_X. Let $f_{\mathcal{H}} : \mathcal{H} \to A$ be the linear form defined on B_X by

(1.3) $$f_{\mathcal{H}} : \mathcal{H} \to A, \qquad f_{\mathcal{H}}(T_{\mathbf{x}}) = \begin{cases} 1 & \text{if } x = 1, \\ 0 & \text{otherwise.} \end{cases}$$

It was shown in [**3**, 2B] that the linear form $f_{\mathcal{H}}$ does not depend on the particular reduced expressions chosen in the definition of the reduced basis B_X. Hence $f_{\mathcal{H}}$ is defined unambiguously on $\mathcal{H}$. Moreover, it was proved in [**3**, Prop. 2.7] that $f_{\mathcal{H}}$ is a trace form. Thus the bilinear form

$$\langle\ ,\ \rangle : \mathcal{H} \otimes \mathcal{H} \to A, \qquad (h_1, h_2) \mapsto f_{\mathcal{H}}(h_1 h_2) \tag{1.4}$$

is symmetric. Clearly the specialization of $\langle\ ,\ \rangle$ to the group algebra of W_n is the standard symmetrizing form which sends (w_1, w_2) to the coefficient of 1 in the product $w_1 w_2$. Since the algebra $\mathcal{H}_K := \mathcal{H}(W_n, \mathbf{u}) \otimes_A K$ over $K = \mathrm{Quot}(A)$ is split semisimple [**2**], this implies:

Theorem 1.5 ([3, Th. 2.8]**).** *The linear form $f_{\mathcal{H}}$ defines a non-degenerate symmetric form*

$$\langle h_1, h_2 \rangle := f_{\mathcal{H}}(h_1 h_2) \qquad \text{for } h_1, h_2 \in \mathcal{H}_K$$

on $\mathcal{H}_K$. This satisfies

$$f_{\mathcal{H}}(T_{\mathbf{x}}) = 0 \quad \text{for all reduced expressions } \mathbf{x} \text{ of non-identity elements of } W_n. \tag{1.6}$$

We will show that $f_{\mathcal{H}}$ actually defines a non-degenerate symmetric form on $\mathcal{H}$ over A. This seems to be important for a further study of this algebra.

Let $\mathcal{O}$ be a domain and $\mathcal{H}$ an unital associative $\mathcal{O}$-algebra. Recall from [**4**, 1C] that an $\mathcal{O}$-basis B of $\mathcal{H}$ with $1 \in B$ is called *quasi-symmetric* if the linear form $f : \mathcal{H} \to \mathcal{O}$ defined by setting $f(b) = \delta_{1b}$ for $b \in B$ is a trace form and via (1.4) gives rise to a non-degenerate symmetric form on $\mathcal{H}$. We then also say that B is *quasi-symmetric with respect to f*. In this terminology, the above theorem can be rephrased to saying that any reduced basis of $\mathcal{H}(W_n, \mathbf{u})_K$ is quasi-symmetric with respect to the form $f_{\mathcal{H}}$ defined above.

2. The Ariki-Koike basis of $\mathcal{H}$ is quasi-symmetric

It will be convenient to introduce a further quasi-symmetric basis of $\mathcal{H} = \mathcal{H}(W_n, \mathbf{u})$. For this let $L_1 := T_0$ and inductively $L_i := q^{-1} T_{i-1} L_{i-1} T_{i-1}$ for $2 \le i \le n$. It was shown by Ariki and Koike [**2**, Th. 3.10] that

$$B := \{ L_1^{b_1} \dots L_n^{b_n} T_w \mid 0 \le b_i \le r-1,\ w \in \mathfrak{S}_n \} \tag{2.1}$$

is a basis of $\mathcal{H}(W_n, \mathbf{u})$. Here T_w denotes the standard basis element in the Iwahori-Hecke subalgebra $\mathcal{H}(\mathfrak{S}_n, q)$ of $\mathcal{H}(W_n, \mathbf{u})$. Note that B is non-reduced if $r > 1$ and $n > 1$.

We can now prove the analogue of (1.6) for this basis.

Proposition 2.2. *The Ariki-Koike basis B is quasi-symmetric with respect to $f_{\mathcal{H}}$, i.e., it satisfies*

$$f_{\mathcal{H}}(b) = 0 \quad \text{for all } 1 \ne b \in B. \tag{2.3}$$

Proof. We compare B to a particular reduced basis B_X of $\mathcal{H}$ and show that any $1 \ne b \in B$ is a linear combination of non-identity basis elements from B_X. The assertion then follows from Theorem 1.5.

To describe B_X we introduce the elements

$$L_{k,a} := \begin{cases} T_{k-1}\dots T_1 T_0^a & \text{for } a>0,\\ 1 & \text{for } a=0,\end{cases}$$

for $1\le k\le n$, $0\le a\le r-1$. According to [**3**, Lemma 1.5] the set

$$B_X := \{L_{1,a_1}\dots L_{n,a_n}T_w \mid 0\le a_i\le r-1,\ w\in\mathfrak{S}_n\}$$

is a reduced basis of $\mathcal{H}$ (the corresponding set X of reduced words is clear). Furthermore, the following relations are proved in [**3**, Lemma 2.3]:

Lemma 2.4. *We have:*

(a)
$$T_iL_{k,a} = \begin{cases} L_{k,a}T_i & \text{if } i>k \text{ or } a=0,\\ L_{k+1,a} & \text{if } i=k, a\neq 0,\\ qL_{k-1,a}+(q-1)L_{k,a} & \text{if } i=k-1, a\neq 0,\\ L_{k,a}T_{i+1} & \text{if } i<k-1, a\neq 0,\end{cases}$$

for $1\le i\le n-1$, $1\le k\le n$.
(b) $L_{k,a}L_{k-m,c} =$

$$L_{k-m-1,c}L_{k,a}T_1 + (q-1)\sum_{i=1}^{c}(L_{k-m-1,a+c-i}L_{k,i} - L_{k-m-1,i}L_{k,a+c-i})$$

for $a,c\neq 0$, $k-m>1$.

Lemma 2.5. *Let* $i\ge 1$.

(a) *Let* $1\le k\le r-1$. *Then* L_i^k *is equal to an* A*-linear combination of basis elements* $L_{1,c_1}\dots L_{i,c_i}T_w\in B_X$ *with* $0\le c_j\le r-1$, $1\le c_i\le k$, $w\in\langle t_1,\dots,t_{i-1}\rangle$.

(b) *Let* $1\le b_i\le r-1$. *Then* $L_1^{b_1}\dots L_i^{b_i}$ *is equal to an* A*-linear combination of basis elements* $L_{1,c_1}\dots L_{i,c_i}T_w\in B_X$ *with* $0\le c_j\le r-1$, $c_i\neq 0$, $w\in\langle t_1,\dots,t_{i-1}\rangle$.

Proof. According to our definitions we have $L_i^k = (q^{1-i}L_{i,1}T_1\dots T_{i-1})^k$. Thus assertion (a) is certainly correct if $k=1$. Now assume that (a) has been proved for all exponents strictly smaller than k. Hence $L_i^k = L_i^{k-1}L_i$ is a linear combination of terms $L_{1,c_1}\dots L_{i,c_i}T_vL_{i,1}T_w$ with $1\le c_i\le k-1$ and $v,w\in\langle t_1,\dots,t_{i-1}\rangle = \mathfrak{S}_i$. But Lemma 2.4(a) shows that $T_vL_{i,1}$ is equal to an A-linear combination of terms $L_{j,1}T_w$ with $j\le i$, $w\in\mathfrak{S}_i$. Thus we have that L_i^k is a linear combination of terms $L_{1,c_1}\dots L_{i,c_i}L_{j,1}T_w$ with $j\le i$, $w\in\mathfrak{S}_i$. Now $L_{i,c_i}L_{j,1}$ with $1\le c_i\le k-1$, $j\le i$ is a linear combination of terms $L_{1,d_1}\dots L_{i,d_i}$ with $1\le d_i\le k$ by Lemma 2.4(b). This shows that L_i^k is a linear combination of terms $hL_{i,d_i}T_w$ with $1\le d_i\le k$ and h in the subalgebra $\mathcal{H}(W_{i-1},\mathbf{u})$.

Expanding h in the reduced basis of this subalgebra we obtain summands of the form $L_{1,c_1}\dots L_{i-1,c_{i-1}}T_vL_{i,c_i}T_w$ with $v\in\langle t_1,\dots,t_{i-2}\rangle$. It remains to apply Lemma 2.4(a) again to move the T_v past L_{i,c_i}. This completes the induction step and hence the proof of assertion (a).

For part (b) we use induction on i. If $i = 1$ then the result is just part (a). So assume that the assertion is proved for all $j < i$. By induction we have that $L_1^{b_1} \dots L_i^{b_i}$ is a linear combination of terms $L_{1,c_1} \dots L_{i-1,c_{i-1}} T_w L_i^{b_i}$ with $w \in \langle t_1, \dots, t_{i-2} \rangle$. By the first part, $L_i^{b_i}$ is an A-linear combination of terms $L_{1,c_1} \dots L_{i,c_i} T_w$ with $c_i \neq 0$. Again the result follows by Lemma 2.4. $\square$

We are now in a position to complete the proof of the proposition. Let $1 \neq b = L_1^{b_1} \dots L_i^{b_i} T_v \in B$ with $a_i \neq 0$. If $i = 0$ then $b = T_v$ is already contained in B_X so there is nothing to prove. If $i > 0$ then according to Lemma 2.5(b) we have that b is equal to a linear combination of basis elements $L_{1,c_1} \dots L_{i,c_i} T_w \in B_X$ with $c_i \neq 0$. In particular, $f_{\mathcal{H}}(b) = 0$ by (1.6). $\square$

3. The determinant of the Gram matrix of type $G(r, 1, n)$

Let $* : \mathcal{H} \longrightarrow \mathcal{H}$ be the A–linear antiautomorphism of $\mathcal{H}$ determined by $T_i^* = T_i$ for $0 \leq i < n$. So, in particular, $L_i^* = L_i$ and $T_w^* = T_{w^{-1}}$ for all i with $1 \leq i \leq n$ and all $w \in \mathfrak{S}_n$. Given a set $H \subseteq \mathcal{H}$ let $H^* = \{ h^* \mid h \in H \}$.

In order to show that $f_{\mathcal{H}}$ induces a symmetric form on $\mathcal{H}$ over A it suffices to exhibit two bases B, C of $\mathcal{H}$ over A such that the determinant of the Gram matrix $\mathcal{G}_{BC} := \big(\langle b, c \rangle\big)_{B,C} = \big(f_{\mathcal{H}}(bc)\big)_{b \in B, c \in C}$ is a unit in A.

For the basis B we take the Ariki–Koike basis defined in (2.1) above. For the basis C we essentially take B^*; however, for technical reasons (see Proposition 3.8 below), the definition is more complicated. Let $\ell_1 = 1$ and for $i = 2, \dots, n$ let $\ell_i = q^{-1} T_{i-1} \ell_{i-1} T_{i-1}$. Then ℓ_i is the analogue of L_i in the subalgebra $\mathcal{H}(\mathfrak{S}_n, q)$; indeed, $L_i = u_1 \ell_i$ if $r = 1$. We define

$$C := \{ T_v \ell_1^{\delta(c_1)} \dots \ell_n^{\delta(c_n)} L_1^{c_1} \dots L_n^{c_n} \mid 0 \leq c_i \leq r - 1,\, v \in \mathfrak{S}_n \}.$$

where $\delta(c) = -1$ if $c \neq 0$ and $\delta(0) = 0$. For a given $c = (c_1, \dots, c_n)$, the set $\{ T_v \ell_1^{\delta(c_1)} \dots \ell_n^{\delta(c_n)} \}_{v \in \mathfrak{S}_n}$ is a basis of $\mathcal{H}(\mathfrak{S}_n, q)$ because each ℓ_i is invertible. Consequently, C is a basis of $\mathcal{H}$ by (2.1).

Theorem 3.1. *Let B and C be as above. Then $\det\big(\mathcal{G}_{BC}\big) = \pm q^l (u_1 \dots u_r)^m$ for some integers l and m.*

By Proposition 2.2, and the above remarks, the main Theorem is a consequence of Theorem 3.1 when W is of type $B_n^{(r)}$; in the next section we deduce the general case from this.

The proof of Theorem 3.1 will occupy the whole of this section. We begin with a well–known Lemma which follows directly from the relations in $\mathcal{H}$ (see, for example, [**2**, Lemma 3.3]).

Lemma 3.2. *Suppose that $1 \leq i < n$ and that $1 \leq j \leq n$. Then*

(a) *L_i commutes with L_j.*
(b) *T_i commutes with L_j if $j \neq i, i+1$.*
(c) *T_i commutes with $L_i L_{i+1}$.*

The elements $\ell_1, \dots, \ell_n$ also satisfy similar relations to those given in the Lemma; however, we shall not need this.

Lemma 3.3. *Suppose that $1 \le i < n$ and let k and m be non–negative integers.*

(a) *If $k \ge m$ then* $L_i^k L_{i-1}^m T_{i-1} = T_{i-1} L_i^m L_{i-1}^k + (q-1) \sum_{c=m+1}^{k} L_i^c L_{i-1}^{k+m-c}$.

(b) *If $k \le m$ then* $L_i^k L_{i-1}^m T_{i-1} = T_{i-1} L_i^m L_{i-1}^k - (q-1) \sum_{c=k+1}^{m} L_i^c L_{i-1}^{k+m-c}$.

Proof. Part (a) is proved in [**6**, Lemma 5.10]; the argument is a routine induction on $k-m$ (use Lemma 3.2(c) to start the induction). To prove (b), apply the involution $*$ to (a) and rearrange. □

Corollary 3.4. *Suppose that $1 < i \le n$ and $k \ge 1$. Then*

$$L_i^k = q^{-1} T_{i-1} L_{i-1}^k T_{i-1} + q^{-1}(q-1) \sum_{c=1}^{k-1} L_i^c L_{i-1}^{k-c} T_{i-1}.$$

Proof. When $k = 1$ this is just the definition of L_i. In general, taking $m = 0$ in part (a) of the Lemma, we see that

$$\begin{aligned} L_i^{k+1} &= q^{-1} L_i^k T_{i-1} L_{i-1} T_{i-1} \\ &= q^{-1} \left\{ T_{i-1} L_{i-1}^k + (q-1) \sum_{c=1}^{k} L_i^c L_{i-1}^{k-c} \right\} L_{i-1} T_{i-1} \\ &= q^{-1} T_{i-1} L_{i-1}^{k+1} T_{i-1} + q^{-1}(q-1) \sum_{c=1}^{k} L_i^c L_{i-1}^{k+1-c} T_{i-1}. \quad \square \end{aligned}$$

Recall the trace form $f_{\mathcal{H}} : \mathcal{H} \longrightarrow A$ from (1.3). Given $h, h' \in \mathcal{H}$ we define an equivalence relation $\overset{f_{\mathcal{H}}}{=}$ on $\mathcal{H}$ by saying that $h \overset{f_{\mathcal{H}}}{=} h'$ whenever $f_{\mathcal{H}}(h) = f_{\mathcal{H}}(h')$. This equivalence relation does not respect multiplication in $\mathcal{H}$; however, because $f_{\mathcal{H}}$ is a trace form, it does have the following crucial property.

(3.5) Let $h, h' \in \mathcal{H}$. Then $hh' \overset{f_{\mathcal{H}}}{=} h'h$.

For $j = 0, \dots, n$ let H_j be the A–submodule of $\mathcal{H}$ with basis

$$\{ L_j^{a_j} \dots L_1^{a_1} T_w \mid 0 \le a_i < r,\ w \in \mathfrak{S}_n \}.$$

Note that $H_n = \mathcal{H}$ and that H_j is not an $\mathcal{H}$–module for $j \ne n$. However, H_j is always a right $\mathcal{H}(\mathfrak{S}_n, q)$–module and, by Proposition 2.2, we also have the next result.

(3.6) Suppose that $1 \le j < n$, $1 \le k \le r-1$ and $h \in H_j$. Then $L_{j+1}^k h \overset{f_{\mathcal{H}}}{=} 0$.

What we really need to know, however, is how to compute $f_{\mathcal{H}}(L_{j+1}^r h)$.

Lemma 3.7. *Suppose that i, j and k are integers such that $1 \leq i \leq j < n$, and $1 \leq k \leq r$ and let $h \in H_j$. Then*

$$L_{j+1}^k h \overset{f_{\mathcal{H}}}{=} q^{i-j-1} T_j \dots T_i L_i^k T_i \dots T_j h.$$

Proof. For the duration of this proof let $T_{j..i} = T_j \dots T_i$ and $T_{i..j} = T_i \dots T_j$. By declaring that $T_{j..i+1} = 1 = T_{i+1..j}$ we can state the lemma when $i = j+1$; in which case the lemma asserts that $f_{\mathcal{H}}(L_{j+1}^k h) = f_{\mathcal{H}}(L_{j+1}^k h)$. Since this is certainly true, we may now proceed by downwards induction on $i \leq j$.

Now, by induction and by Corollary 3.4,

$$\begin{aligned} q^{j-i+2} L_{j+1}^k h &\overset{f_{\mathcal{H}}}{=} q T_{j..i} L_i^k T_{i..j} h \\ &= T_{j..i} T_{i-1} L_{i-1}^k T_{i-1} T_{i..j} h + (q-1) \sum_{c=1}^{k-1} T_{j..i} L_i^c L_{i-1}^{k-c} T_{i-1} T_{i..j} h. \end{aligned}$$

Thus, it suffices to show that $T_{j..i} L_i^c L_{i-1}^{k-c} T_{i-1} T_{i..j} h \overset{f_{\mathcal{H}}}{=} 0$ whenever $0 < c < k$. Fix such an integer c. Then, using the definition of the elements L_a together with Lemma 3.2, we find

$$\begin{aligned} T_{j..i} L_i^c L_{i-1}^{k-c} T_{i-1} T_{i..j} h &= q^{j-i+2} T_{j..i} L_i^c L_{i-1}^{k-c-1} \big(T_{i-1}^{-1} T_{j..i}^{-1} L_{j+1}\big) h \\ &= q^{j-i+2} T_{j..i} \big(L_{i-1}^{k-c-1} L_i^c\big) T_{i-1}^{-1} T_{j..i}^{-1} L_{j+1} h \\ &= q^{j-i+1} T_{j..i} L_{i-1}^{k-c-1} \big(L_i^{c-1} T_{i-1} L_{i-1}\big) T_{j..i}^{-1} L_{j+1} h \\ &= q^{j-i+1} T_{j..i} L_i^{c-1} L_{i-1}^{k-c-1} T_{i-1} T_{j..i}^{-1} L_{j+1} L_{i-1} h. \end{aligned}$$

Now $T_a^{-1} = q^{-1}(T_a + 1 - q)$ for $1 \leq a < n$. Therefore, by Lemma 3.3, together with Lemma 3.2(a), there exist scalars $a_L \in A$ such that

$$L_i^{c-1} L_{i-1}^{k-c-1} T_{i-1} T_{j..i}^{-1} = \sum_{L = T_w L_{j+1}^{b_{j+1}} \dots L_1^{b_1} \in B^*} a_L T_w L_{j+1}^{b_{j+1}} \dots L_1^{b_1}$$

where $b_{j+1} \leq \max\{c-1, k-c-1\} \leq k-2$ whenever $a_L \neq 0$. Consequently,

$$\begin{aligned} T_{j..i} L_i^c L_{i-1}^{k-c} T_{i-1} T_{i..j} h &= \sum_{L = T_w L_{j+1}^{b_{j+1}} \dots L_1^{b_1}} q^{j-i+1} a_L T_{j..i} T_w L_{j+1}^{b_{j+1}} \dots L_1^{b_1} L_{j+1} L_{i-1} h \\ &\overset{f_{\mathcal{H}}}{=} \sum_{L = T_w L_{j+1}^{b_{j+1}} \dots L_1^{b_1}} q^{j-i+1} a_L L_{j+1}^{1+b_{j+1}} \dots L_1^{b_1} L_{i-1} h T_{j..i} T_w \end{aligned}$$

by (3.5). However, $1 + b_{j+1} \leq k-1 < r$; so $T_{j..i} L_i^c L_{i-1}^{k-c} T_{i-1} T_{i..j} h \overset{f_{\mathcal{H}}}{=} 0$ by (3.6) as required. □

Recall the element $\ell_{j+1} = q^{-j} T_j \dots T_1 T_1 \dots T_j$.

Proposition 3.8. *Let j be an integer such that $1 \le j < n$ and suppose that $h \in H_j$. Then*

$$L_{j+1}^r h \overset{f_{\mathcal{H}}}{=} (-1)^r u_1 \dots u_r h \ell_{j+1}.$$

Proof. By Lemma 3.7, $f_{\mathcal{H}}(L_{j+1}^r h) = q^{-j} f_{\mathcal{H}}(T_j \dots T_1 T_0^r T_1 \dots T_j h)$. Therefore, since $(T_0 - u_1) \dots (T_0 - u_r) = 0$, we may write $f_{\mathcal{H}}(L_{j+1}^r h)$ as an A–linear combination of terms $f_{\mathcal{H}}\big(T_j \dots T_1 T_0^k T_1 \dots T_j h\big)$ for $k = 0, \dots, r-1$. However, by Lemma 3.7 and (3.6), if $0 < k < r$ then

$$T_j \dots T_1 T_0^k T_1 \dots T_j h \overset{f_{\mathcal{H}}}{=} q^j L_{j+1}^k h \overset{f_{\mathcal{H}}}{=} 0.$$

Therefore, $L_{j+1}^r h \overset{f_{\mathcal{H}}}{=} (-1)^r q^{-j} u_1 \dots u_r h T_j \dots T_1 T_1 \dots T_j = (-1)^r u_1 \dots u_r h \ell_{j+1}$ as required. □

We can now prove the main result of this section.

Proof of Theorem 3.1. Recall the bases B and C of $\mathcal{H}$ and the Gram matrix $\mathcal{G}_{BC} = \big(f_{\mathcal{H}}(bc)\big)_{b \in B, c \in C}$ from the beginning of this section.

An arbitrary element of B is of the form $L_n^{b_n} \dots L_1^{b_1} T_w$ for some $w \in \mathfrak{S}_n$ and some integers $0 \le b_i < r$. In the matrix $\mathcal{G}_{BC}$ order the elements of B lexicographically according to the exponents $(b_n, \dots, b_1)$, and similarly for the elements of C.

For $j = 0, \dots, n$ let $B_j = B \cap H_j$ and $C_j = C \cap H_j^*$ and define $\mathcal{G}_j$ to be the Gram matrix $\mathcal{G}_{B_j C_j} = \big(f_{\mathcal{H}}(bc)\big)_{b \in B_j, c \in C_j}$. Then $\mathcal{G}_{BC} = \mathcal{G}_n$ and our ordering of the elements of B and C is such that the matrix $\mathcal{G}_j$ sits in the top left hand corner of the matrix $\mathcal{G}_{j+1}$ for each $0 \le j < n$.

Suppose that $0 \le j < n$ and let $\alpha = (-1)^r u_1 \dots u_r$. Then we claim that, as an $r \times r$–block matrix, $\mathcal{G}_{j+1}$ has the form

$$\begin{pmatrix} \mathcal{G}_j & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & \alpha \mathcal{G}_j \\ \vdots & \vdots & \iddots & \iddots & * \\ 0 & 0 & \iddots & * & \vdots \\ 0 & \alpha \mathcal{G}_j & * & \cdots & * \end{pmatrix}$$

To see this let $b = L_{j+1}^{b_{j+1}} \dots L_1^{b_1} T_w$ and $c = T_v \ell_1^{\delta(c_1)} \dots \ell_{j+1}^{\delta(c_{j+1})} L_1^{c_1} \dots L_{j+1}^{c_{j+1}}$ be arbitrary elements of B_{j+1} and C_{j+1} respectively. Then

$$\begin{aligned} bc &= L_{j+1}^{b_{j+1}} \dots L_1^{b_1} T_w T_v \ell_1^{\delta(c_1)} \dots \ell_{j+1}^{\delta(c_{j+1})} L_1^{c_1} \dots L_{j+1}^{c_{j+1}} \\ &\overset{f_{\mathcal{H}}}{=} L_{j+1}^{b_{j+1}+c_{j+1}} L_j^{b_j+c_j} \dots L_1^{b_1+c_1} T_w T_v \ell_1^{\delta(c_1)} \dots \ell_j^{\delta(c_j)} \ell_{j+1}^{\delta(c_{j+1})} \end{aligned}$$

by (3.5) and Lemma 3.2(a). If $b_{j+1} = c_{j+1} = 0$ the assertion is just the definition of $\mathcal{G}_j$ (note that $\delta(0) = 0$). If $0 < b_{j+1} + c_{j+1} < r$ then $f_{\mathcal{H}}(bc) = 0$ by (3.6). Finally, if $b_{j+1} + c_{j+1} = r$ then by Proposition 3.8

$$\begin{aligned} bc &\overset{f_{\mathcal{H}}}{=} (-1)^r u_1 \dots u_r L_j^{b_j+c_j} \dots L_1^{b_1+c_1} T_w T_v \ell_1^{\delta(c_1)} \dots \ell_j^{\delta(c_j)} \ell_{j+1}^{\delta(c_{j+1})} \ell_{j+1} \\ &= (-1)^r u_1 \dots u_r L_j^{b_j+c_j} \dots L_1^{b_1+c_1} T_w T_v \ell_1^{\delta(c_1)} \dots \ell_j^{\delta(c_j)}, \end{aligned}$$

since $\delta(c_{j+1}) = -1$ because $c_{j+1} > 0$. This proves the claim.

Therefore, $\det \mathcal{G}_{j+1} = \pm(u_1 \dots u_r)^{r-1} \big(\det \mathcal{G}_j\big)^r$ for all $0 \le j < n$. Consequently, $\det \mathcal{G}_{BC} = \det \mathcal{G}_n = \pm(u_1 \dots u_r)^m \big(\det \mathcal{G}_0\big)^k$ for some integers k and m. However, $\mathcal{G}_0 = \big(f_{\mathcal{H}}(T_w T_v)\big)_{v,w \in \mathfrak{S}_n}$ and it is well–known, and easy to prove, that $f_{\mathcal{H}}(T_w T_v) = q^l \delta_{wv^{-1}}$, where l is the length of a reduced expression for w. Hence, $\det \mathcal{G}_0 = \pm q^z$ for some integer z. This completes the proof of Theorem 3.1. $\square$

4. The determinant of the Gram matrix of type $G(r,2,2)$

For any divisor p of r let $W_{n,p}$ be the normal subgroup of index p of W_n generated by the $n+1$ reflections

$$\{t_0^p, t_0^{-1} t_1 t_0, t_1, \dots, t_{n-1}\}$$

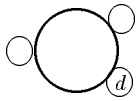
if $p < r$, respectively the n reflections

$$\{t_0^{-1} t_1 t_0, t_1, \dots, t_{n-1}\}$$

if $p = r$. Thus $W_{n,p}$ is the imprimitive complex reflection group $G(r,p,n)$ in the notation of Shephard and Todd.

We would like to introduce a symmetrizing form on the generic cyclotomic algebra $\mathcal{H}(W_{n,p}, \mathbf{u})$ defined in [**5**]. This coincides with the algebra introduced by Ariki in [**1**] except for the case where $n = 2$, and both r and p are even, where the generic algebra has one additional parameter. In this latter case, no quasi-symmetric basis had been given in [**3**], so we need to treat this algebra separately.

So, suppose that r is even, $n = p = 2$, and set $d := r/2$. Then $W_{2,2}$ has a presentation described by the diagram

(4.1) $$D_2^{(d)}: \quad$$ 

Let $\mathcal{H} = \mathcal{H}(W_{2,2}, \mathbf{u})$, where $\mathbf{u} = (u_1, \dots, u_d; q; q')$, be the corresponding cyclotomic algebra, generated as associative unital $\mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$-algebra by elements R, S, T subject to the relations

$$\begin{aligned} &(T-u_1)\dots(T-u_d) = 0 = (R-q)(R+1) = (S-q')(S+1), \\ &RST = STR = TRS \end{aligned}$$

(see [**4**, 4A]). We let $U := RS$ and $V := \frac{q'-1}{q'} U + q - 1$. Note that T and U commute and the element TU is central in $\mathcal{H}$. The following result is inspired by [**1**, Lemma 1.2].

Lemma 4.2. *Let j, k be nonnegative integers. Then*

(a) $$ST^j = (qq')^{-j} T^j U^{2j} S - \tfrac{1}{q} T^j U V \sum_{i=0}^{j-1} (qq')^{-i} U^{2i},$$

(b) $$SU^k = (qq')^k U^{-k} S + \tfrac{1}{q} U V \sum_{i=0}^{k-1} (qq')^{k-i} U^{2i-k},$$

(c) $ST^jU^k = (qq')^{k-j}T^jU^{2j-k}S + \begin{cases} -\frac{1}{q}T^jV\sum_{i=0}^{j-k-1}(qq')^{-i}U^{2i+1+k} & \text{if } j \geq k, \\ \frac{1}{q}T^jV\sum_{i=j-k}^{-1}(qq')^{-i}U^{2i+1+k} & \text{if } j < k. \end{cases}$

Proof. We prove (a) by induction on j. The result trivially holds if $j = 0$. If $j = 1$ then

$$\begin{aligned} ST = STRR^{-1} &= \frac{1}{q}TUR - \frac{q-1}{q}TU = \frac{1}{q}TU^2S^{-1} - \frac{q-1}{q}TU \\ &= (qq')^{-1}TU^2S - \frac{1}{q}TUV, \end{aligned}$$

as claimed. The induction hypothesis and the case $j = 1$ now yield

$$\begin{aligned} ST^{j+1} &= (qq')^{-j}T^jU^{2j}ST - \frac{1}{q}T^{j+1}U\sum_{i=0}^{j-1}(qq')^{-i}U^{2i}V \\ &= (qq')^{-j-1}T^{j+1}U^{2j+2}S - \frac{1}{q}T^{j+1}U\sum_{i=0}^{j}(qq')^{-i}U^{2i}V, \end{aligned}$$

proving the first part.

In part (b) the case $k = 0$ is again obvious. If $k = 1$ then

$$\begin{aligned} SU = SRS &= (q'S^{-1} + q' - 1)(qR^{-1} + q - 1)S \\ &= qq'U^{-1}S + (q-1)q' + (q'-1)(qR^{-1} + q - 1)S \\ &= qq'U^{-1}S + (q-1)q' + (q'-1)U = qq'U^{-1}S + q'V. \end{aligned}$$

The inductive step is verified by the same type of calculation as in the first part.

The last part is an immediate consequence of parts (a) and (b). □

Note that T, U generate a commutative subalgebra of $\mathcal{H}$. Since it is known that $\mathcal{H}$ is $\mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$-free of rank $|W_{2,2}|$ by [**4**, Satz 4.7] we conclude as there that

$$\{T^jU^k,\ T^jU^kS \mid 0 \leq 2j - k, k \leq r - 1\}$$

is a $\mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$-basis of $\mathcal{H}$ (compare also with [**1**, Prop. 1.4] where the case $q = q'$ is considered).

We proceed to show that this basis has good properties.

Proposition 4.3. *The basis*

$$C = \{T^jU^k,\ T^jU^kS \mid 0 \leq 2j - k, k \leq r - 1\} \tag{4.4}$$

of $\mathcal{H} = \mathcal{H}(W_{2,2}, \mathbf{u})$ *is quasi-symmetric.*

Proof. We have to show that the linear form $f_{\mathcal{H}} : \mathcal{H} \to \mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$ defined by $f_{\mathcal{H}}(1) = 1$, $f_{\mathcal{H}}(b) = 0$ if $1 \neq b \in C$ is a trace form. Since it specializes to the standard symmetrizing form of the complex reflection group $W_{2,2}$ the induced symmetric form on $\mathcal{H}$ is then automatically non-degenerate over the quotient field of A.

We will have to consider elements of $\mathcal{H}$ of the form $T^jU^kS^l$ with $0 \le k \le r-1$, $l \in \{0,1\}$ and j an arbitrary integer. First note that on such elements

$$f_{\mathcal{H}}(T^jU^kS^l) = 0 \qquad \text{if } k \neq 0 \text{ or } l \neq 0. \tag{4.5}$$

Indeed, if $T^jU^kS^l \notin C$ then by the relation on T we may replace T^j by a linear combination of elements T^{j+i} or T^{j-i}, with $1 \le i \le d$. Induction and the definition of $f_{\mathcal{H}}$ now show the claim.

Since R and S are invertible in $\mathcal{H}$ it clearly suffices to prove that $f_{\mathcal{H}}(bL) = f_{\mathcal{H}}(Lb)$ for $b \in C$ and $L \in \{S, U, TU\}$. This is immediate for the central element $L = TU$. Now assume $L = S$. Inspection shows that if $0 \le 2j-k, k \le r-1$ then all monomials on the right hand side of Lemma 4.2(c) lie in C. Thus, Lemma 4.2(c) gives the decomposition in C of ST^jU^k. By the definition of $f_{\mathcal{H}}$ it follows that $f_{\mathcal{H}}(S \cdot T^jU^k) = 0 = f_{\mathcal{H}}(T^jU^k \cdot S)$. Similarly, $f_{\mathcal{H}}(T^jU^kS \cdot S)$ equals

$$f_{\mathcal{H}}(S \cdot T^jU^kS) = \begin{cases} q' & \text{if } j = k = 0, \\ 0 & \text{else.} \end{cases}$$

So finally assume that $L = U$. As U commutes with T, the assertion is again clear if $b = T^jU^k$. For $b = T^jU^kS$ we have to compare the value of $f_{\mathcal{H}}$ on $T^jU^{k+1}S$ and on

$$T^jU^kSU = qq'T^jU^{k-1}S + q'(q-1)T^jU^k + (q'-1)T^jU^{k+1} \tag{4.6}$$

(by Lemma 4.2(b)). If $k \neq 0, r-1$ and $2j-k \neq 0, r-1$ then all monomials in the above expressions lie in C and $f_{\mathcal{H}}$ vanishes on both sides. If $2j-k \in \{0, r-1\}$ then $f_{\mathcal{H}}$ again vanishes by (4.5). We consider the two remaining cases separately.

First assume that $k = 0$. We have $f_{\mathcal{H}}(U \cdot T^jS) = f_{\mathcal{H}}(ST^jU)$ by what we proved for S, so the result is true if $j = 0$. For $j > 0$

$$ST^jU = TUST^{j-1} = (qq')^{1-j}T^jU^{2j-1}S - \frac{1}{q}T^jU^2V\sum_{i=0}^{j-2}(qq')^{-i}U^{2i}$$

by Lemma 4.2(a). All monomials appearing in the right-hand side are non-trivial basis elements, so $f_{\mathcal{H}}$ vanishes. On the other hand we have the equality

$$\begin{aligned} SU &= S\sum_{l=0}^{d-1} a_lT^{l+1}U = \sum_{l=0}^{d-1} a_lTUST^l \\ &= \sum_{l=0}^{d-1} a_l\left((qq')^{-l}T^{l+1}U^{2l+1}S - \frac{1}{q}T^{l+1}U^2V\sum_{i=0}^{l-1}(qq')^{-i}U^{2i}\right) \end{aligned}$$

(with suitable constants a_l) since TU is central and by Lemma 4.2(a). Here, $f_{\mathcal{H}}$ is zero by (4.5).

It remains to consider $k = r-1$. Lemma 4.2(a) gives

$$\begin{aligned} ST^d =& (qq')^{-d}T^dU^r(S-(q'-1)) \\ &- \frac{q-1}{q}T^dU^{r-1}(qq')^{1-d} - \frac{1}{q}T^dUV\sum_{i=0}^{d-2}(qq')^{-i}U^{2i}. \end{aligned} \tag{4.7}$$

The last two terms on the right-hand side are of the type (4.5). On the other hand, replacing T^d by a linear combination of T^i with $0 \le i \le d-1$ and again applying Lemma 4.2(a) shows that the left-hand side is a sum of elements of the form considered in (4.5). Multiplication of both sides by $T^{-(d-j)}$ does not change this fact, and thus $f_{\mathcal{H}}(T^jU^rS) = (q'-1)f_{\mathcal{H}}(T^jU^r)$. But now (4.6), with $k = r-1$, gives

$$f_{\mathcal{H}}(T^jU^{r-1}S \cdot U) = (q'-1)f_{\mathcal{H}}(T^jU^r) = f_{\mathcal{H}}(T^jU^rS) = f_{\mathcal{H}}(U \cdot T^jU^{r-1}S)$$

as required. □

We write $\mathcal{G}_C$ for the Gram matrix of $f_{\mathcal{H}}$ with respect to the pair of bases C, C^*, where C^* is obtained from C by reversing the order of the factors in each element.

Proposition 4.8. *Let C be the basis of $\mathcal{H}$ defined in (4.4). Then* $\det(\mathcal{G}_C)$ *is invertible in $A = \mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$. In particular $\mathcal{H}$ is symmetric over A.*

Proof. Elements of C are of the form $T^jU^kS^l$ for some j, k, l with $0 \le k, 2j-k \le r-1$ and $l \in \{0,1\}$. We order C lexicographically with respect to (k, l, j). Then (4.5) shows that the Gram matrix has the shape

$$\begin{pmatrix} \mathcal{G}_1 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & \mathcal{G}_2 \\ \vdots & \vdots & \iddots & \iddots & * \\ 0 & 0 & \iddots & * & \vdots \\ 0 & \mathcal{G}_r & * & \cdots & * \end{pmatrix}$$

(as an $r \times r$-block matrix). Again by (4.5) we see that $\mathcal{G}_1$ is a block diagonal 2×2-matrix, and the blocks on the diagonal are the same as for the algebra $\mathcal{H}(W_1, \mathbf{u})$. Thus the determinant of $\mathcal{G}_1$ has the required form. The matrices $\mathcal{G}_i$, $2 \le i \le r$, can also be considered as 2×2-block matrices, with entries $f_{\mathcal{H}}(T^{j_1+j_2}U^rS^{l_1+l_2})$ where $l_1, l_2 \in \{0,1\}$. Application of (4.7) shows that $\mathcal{G}_i$ has the block form

$$\begin{pmatrix} \mathcal{G}'_i & (q'-1)\mathcal{G}'_i \\ (q'-1)\mathcal{G}'_i & (q'+(q'-1)^2)\mathcal{G}'_i \end{pmatrix}.$$

Here $\mathcal{G}'_i$ is the matrix with entries $f_{\mathcal{H}}(T^{j_1+j_2}U^r)$. Subtraction of $(q'-1)$ times the first block of r columns from the second block of r columns thus yields $\det \mathcal{G}_i = q' \det(\mathcal{G}'_i)^2$. To evaluate the determinant of $\mathcal{G}'_i$ we need a further calculation. Application of (4.7) gives

$$\begin{aligned} T^lU^r &= \frac{1}{q'}T^lU^r(S-(q'-1))S \\ &= \frac{1}{q'}(qq')^dT^{l-d}\left(ST^d + \frac{q-1}{q}T^dU^{r-1}(qq')^{1-d} - \frac{1}{q}T^dUV\sum_{i=0}^{d-2}(qq')^{-i}U^{2i}\right)S \\ &= q(qq')^{d-1}T^{l-d}ST^dS + (q-1)T^lU^{r-1}S - T^lUV\sum_{i=0}^{d-2}(qq')^{-i-1}U^{2i}S. \end{aligned}$$

According to (4.5) the form $f_{\mathcal{H}}$ vanishes on all summands on the right-hand side, except possibly the first. Hence, expanding $T^d = \sum_{j=0}^{d-1} a_j T^j$, we have

$$\begin{aligned} T^l U^r &\overset{f_{\mathcal{H}}}{=} q(qq')^{d-1} T^{l-d} S \sum_{j=0}^{d-1} a_j T^j S \\ &= q(qq')^{d-1} T^{l-d} \sum_{j=0}^{d-1} a_j \left((qq')^{-j} T^j U^{2j} S - \frac{1}{q} T^j U V \sum_{i=0}^{j-1} (qq')^{-i} U^{2i} \right) S \\ &\overset{f_{\mathcal{H}}}{=} q(qq')^{d-1} T^{l-d} a_0 S^2 \overset{f_{\mathcal{H}}}{=} (qq')^d u_1 \dots u_d T^{l-d} \end{aligned}$$

by Lemma 4.2(a) and (4.5). Note that this is the analogue of Proposition 3.8 in this case. In particular this last equation shows that the determinant of $\mathcal{G}_i'$ is the same as the one occurring in $\mathcal{H}(W_1, \mathbf{u})$ (up to units in A). The assertion follows. □

Let $f_{\mathcal{H}} : \mathcal{H} \to A$ be the symmetrizing form induced by the quasi-symmetric basis C. Over the quotient field $K := \mathrm{Quot}(A)$ we may decompose $f_{\mathcal{H}}$ as a sum

$$f_{\mathcal{H}} = \sum_{\chi \in \mathrm{Irr}(\mathcal{H}_K)} \delta_\chi \, \chi$$

of irreducible characters of $\mathcal{H}_K := \mathcal{H} \otimes K$, with $\delta_\chi \in K$. The coefficients δ_χ are called the *relative degrees* associated to $f_{\mathcal{H}}$, respectively to the quasi-symmetric basis C. Clearly we have

$$\sum_{\chi \in \mathrm{Irr}(\mathcal{H}_K)} \delta_\chi \, \chi(c) = 0$$

for all elements $c \neq 1$ of a quasi-symmetric basis with respect to $f_{\mathcal{H}}$, and $\sum \delta_\chi \, \chi(1) = 1$. Using this property of the relative degrees the first author in [**8**] computed candidates for the relative degrees attached to a nice symmetrizing form on $\mathcal{H}$. We can now verify:

Corollary 4.9. *The relative degrees of the symmetric form induced by the quasi-symmetric basis C are those given in* [**8**, Th. 3.9]

Proof. According to the result in [**8**] it suffices to show that $f_{\mathcal{H}}$ vanishes on the elements

$$\{RT^jU^j,\ ST^jU^j \mid 0 \leq j \leq d-1\} \cup \{T^{i+j}U^j \mid 0 < 2i + j \leq r-1\}$$

of $\mathcal{H}$. But $RT^jU^j = T^jU^{j+1}S^{-1} = \frac{1}{q'} T^jU^{j+1}(S - (q'-1))$, on which $f_{\mathcal{H}}$ vanishes by (4.7), $ST^jU^j = T^jU^jS \in C \setminus \{1\}$, and also $T^{i+j}U^j \in C$, so the result follows. □

5. The cyclotomic algebras $\mathcal{H}(G(r,p,n), \mathbf{u})$

Finally, we consider the general case where $W_{n,p}$ is an arbitrary imprimitive complex reflection group $G(r,p,n)$. Let $\mathcal{H}(W_{n,p}, \mathbf{u})$ be the generic cyclotomic algebra associated to $W_{n,p}$ in [**5**]. This coincides with the algebra defined by Ariki [**1**, Prop. 1.6] except in the case where $n = 2$ and r, p are both even.

Theorem 5.1. *Let $W = G(r,p,n)$ be an imprimitive complex reflection group. Then the associated generic cyclotomic algebra $\mathcal{H} = \mathcal{H}(W, \mathbf{u})$ carries a canonical symmetrizing form $f_{\mathcal{H}}$ over $\mathbb{Z}[\mathbf{u}, \mathbf{u}^{-1}]$.*

This form vanishes on all non-identity elements of the Ariki-Koike basis of $\mathcal{H}$, while $f_{\mathcal{H}}(1) = 1$. In the case $p = 1$ it also vanishes on all non-identity elements of any reduced basis of $\mathcal{H}$.

Proof. We reduce this to the case $p = 1$ (respectively $p = 2$ if $n = 2$ and r, p are even) where such a symmetrizing form was found in Theorem 3.1 (resp. Proposition 4.8). Assume first that either $n > 2$ or one of r or p is odd. Let $d := r/p$, $\zeta_p := \exp(2\pi\mathrm{i}/p)$ and $\tilde{A} := \mathbb{Z}[\zeta_p, \mathbf{u}, \mathbf{u}^{-1}]$. Then according to Ariki [**1**, Prop. 1.6] the algebra $\mathcal{H} \otimes \tilde{A}$ can be identified with a subalgebra of the specialized $\tilde{A}$–algebra $\tilde{\mathcal{H}} = \mathcal{H}(W_n, \tilde{\mathbf{u}})$, with parameters

$$\tilde{\mathbf{u}} = (u_1, \dots, u_d, \zeta_p u_1, \dots, \zeta_p u_d, \dots, \zeta_p^{p-1} u_1, \dots, \zeta_p^{p-1} u_d; q)\,, \tag{5.2}$$

generated by

$$\{T_0^p,\ T_0^{-1} T_1 T_0,\ T_1,\ \dots,\ T_{n-1}\}\,.$$

Let $\tilde{f}$ be the linear form on $\tilde{\mathcal{H}}$ obtained by specializing the symmetrizing form constructed in Section 3. Since the specialization to the group algebra factorizes through the specialization (5.2) the form $\tilde{f}$ gives rise to a non-degenerate symmetric form on $\tilde{\mathcal{H}}$. In particular, all irreducible characters of $\tilde{\mathcal{H}} \otimes \mathbb{Q}(\mathbf{u})$ occur with non-zero multiplicity in $\tilde{f}$. This implies that all irreducible characters of the subalgebra $\mathcal{H} \otimes \tilde{A}$ of $\tilde{\mathcal{H}}$ occur with non-zero multiplicity in the restriction $f_{\mathcal{H}}$ of $\tilde{f}$ to $\mathcal{H} \otimes \tilde{A}$. Thus, $f_{\mathcal{H}}$ is a symmetrizing form for $\mathcal{H}$ over the quotient field of $\tilde{A}$.

It is shown in [**1**, Prop. 1.6] that

$$\{L_n^{a_n} \dots L_1^{a_1} T_w \mid 0 \le a_i \le r-1,\ \textstyle\sum a_i \equiv 0 \pmod p,\ w \in \mathfrak{S}_n\}$$

is an A-basis of $\mathcal{H}$. Tensoring with $\tilde{A}$, this gives a basis of $\mathcal{H} \otimes \tilde{A}$ which is a subset of the Ariki-Koike basis of the cyclotomic algebra $\tilde{\mathcal{H}}$; it follows that $f_{\mathcal{H}}$ is a symmetrizing form for $\mathcal{H} \otimes \tilde{A}$. However, the Gram determinants for the bases of $\mathcal{H}$ and $\mathcal{H} \otimes \tilde{A}$ are equal and are elements of A, so $f_{\mathcal{H}}$ is a symmetrizing form for $\mathcal{H}$ (over A), as required.

In the case where $n = 2$ and r and p are even, we embed $\mathcal{H}$ instead into a suitable specialization $\mathcal{H}(W_{2,2}, \tilde{\mathbf{u}})$ of the generic algebra of $W_{2,2}$ studied in 4.A (see [**8**, Prop. 3.8]). The remainder of the argument is then the same as in the general case. □

Remark 5.3. We expect that $f_{\mathcal{H}}$ should also vanish on non-identity reduced basis elements in the general case $p \neq 1$. In this context note that a word which is reduced in $W_{n,p}$ does not necessarily have to be reduced in W_n.

Remark 5.4. In [**7**, Verm. 2.20 and Satz 5.13] the first author put forward a conjecture on the values of the relative degrees δ_χ associated to a nice symmetrizing form on $\mathcal{H}(W_{n,p})$. Using the calculations in [**8**, Th. 3.4] and Corollary 4.9 it follows that this conjecture is correct if $n = 2$ for the symmetrizing form studied in this paper. We expect that in general the relative degrees attached to the symmetrizing form $f_{\mathcal{H}}$ above are those predicted in [**7**].

Finally, suppose that $\mathcal{O}$ is an integral domain and let $\mathcal{H}_{\mathcal{O}}(W_{n,p}, \hat{\mathbf{u}})$ be the specialized cyclotomic Hecke algebra of $W_{n,p}$ with parameters $\hat{\mathbf{u}} \in \mathcal{O}$. Given a set of indeterminates $\mathbf{u}$ over $\mathbb{Z}$ we can consider $\mathcal{O}$ as a $\mathbb{Z}[\mathbf{u}]$–module by specifying that $\mathbf{u}$ acts as $\hat{\mathbf{u}}$. Then $\mathcal{H}_{\mathcal{O}}(W_{n,p}, \hat{\mathbf{u}}) \cong \mathcal{H}_{\mathbb{Z}[\mathbf{u}]}(W_{n,p}, \mathbf{u}) \otimes_{\mathbb{Z}[\mathbf{u}]} \mathcal{O}$. Therefore, by Theorem 5.1, we have the following.

Corollary 5.5. *Let $\mathcal{H}_{\mathcal{O}}(W_{n,p}, \hat{\mathbf{u}})$ be a specialized cyclotomic Hecke algebra over $\mathcal{O}$. Then $\mathcal{H}_{\mathcal{O}}(W_{n,p}, \hat{\mathbf{u}})$ is a symmetric algebra with respect to $f_{\mathcal{H}}$ if and only if the parameters $\hat{\mathbf{u}}$ are invertible in $\mathcal{O}$.*

In [**6**] an analogue of the $\hat{q}$–Schur algebra for a cyclotomic Hecke algebra $\mathcal{H}_{\mathcal{O}}$ of type $B_n^{(r)}$ was introduced. Key to the main results of that paper is the calculation of the "double annihilator" of certain elements of $\mathcal{H}$ (see [**6**, Thm. 5.16]). When the parameters $\mathbf{u}$ are invertible elements of $\mathcal{O}$, the required result of [**6**] is an immediate consequence of Corollary 5.5 (significantly simplifying the argument there).

Interestingly, the results of [**6**] require only that $\hat{q}$ be invertible in $\mathcal{O}$. This suggests that $\mathcal{H}_{\mathcal{O}}$ may be a quasi–Frobenius algebra so long as $\hat{q}$ is an invertible element of $\mathcal{O}$.

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